

Spontaneous population activity reconstructs the orientation-preference map in a subset of macaque V1 arrays

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Abstract

Cortical circuits are thought to replay their functional architecture in spontaneous activity, but whether a fine-grained sensory map can be estimated from resting-state population activity *alone*, without any stimulus-evoked reference beyond the map itself, is unclear. We asked whether the orientation-preference (OP) map of macaque V1, measured with moving-bar stimulation, can be reconstructed from the correlation structure of spontaneous multi-unit activity (tMUA) recorded on chronic Utah arrays. For each array we computed the channel-by-channel correlation matrix during eyes-closed, eyes-open, and pooled resting-state activity, reduced it with PCA, and read out each channel’s derived orientation from its polar angle in each pair of principal components, allowing an optimal rotation and mirror flip against the real map. Significance was assessed against a uniform-random control (2000 shuffles per array) using a selection-corrected minimum-over-pairs test, which is markedly stricter than a naive per-pair test. Across 3 monkeys, 29 arrays, and up to 3 recording days, a subset of arrays reconstructed the OP map far better than the control (6/29 arrays were estimated in a majority of their day $\times$ condition cells; 16/29 showed significant estimation in at least one cell). Estimation was reproducible: the same arrays (monkey L: arrays 13, 14, 16; monkey N: arrays 1, 6) were estimated on independent recording days against freshly redrawn controls. Estimation required sufficient sampling: arrays significant at the 5% level had systematically more oriented channels, though the count alone did not guarantee estimation. The principal-component pair that best carried the map was not fixed across arrays or conditions, and eyes-closed/eyes-open estimation, while correlated, diverged in a substantial minority of cases. These results indicate that the correlation structure of spontaneous V1 activity encodes orientation-map information that a purely geometric, unsupervised readout can estimate, but only where the local population is sampled densely enough — a necessary constraint for any attempt to infer functional architecture from resting state alone.

1 Introduction

(placeholder — background on spontaneous activity reflecting the functional architecture of sensory cortex, prior evidence in visual cortex (e.g. imaging-based work on spontaneous OP-map replay), the specific novelty here: Utah-array tMUA rather than imaging, a fully geometric/unsupervised readout, and a selection-corrected significance test. State the question directly: can PCA on the spontaneous correlation matrix alone estimate a previously measured OP map, and for which arrays/conditions does this work?)

2 Methods

Animals and recordings. Three macaques (L, N, F) implanted with 64-channel V1 Utah arrays (L: 14 arrays over 3 resting-state recording days; N: 7 arrays over 2 days; F: 8 arrays over 2 days).

Each resting-state recording was split into eyes-closed (EC) and eyes-open (EO) epochs based on an independent eye-state indicator; results are also reported for the pooled recording (“all”).

Firing-rate matrix. Thresholded multi-unit activity (tMUA) at 1 ms resolution was binned into a firing-rate matrix per channel. Bin size was swept over $\{20, 50, 100, 250, 500, 750, 1000\}$ ms (Fig. S1); 100 ms gave the highest fraction of significant cells and was used throughout the main figures. No binarization or up-state detection was applied to the firing rate — this is a deliberate simplification relative to earlier pipelines in this project.

Channel selection. A channel was excluded from the analysis (marked *unoriented*) if its selectivity index (`selectivity_01`) was below 0.15, or if it showed more than 2 large jumps in its tuning curve (`num_f0_high_jump` > 2). This selection is applied identically to the real map and to every control map.

Correlation and PCA. For the oriented channels only, we computed the full Pearson correlation matrix of the firing-rate traces and performed PCA on it (up to 7 components; “Option A” — PCA was not run on the full 64-channel matrix, which was tried and dropped in favour of restricting to oriented channels).

Geometric OP readout. For each of the $\binom{7}{2} = 21$ pairs of principal components, each channel’s polar angle around the centroid of the PC-score cloud was computed, then rotated (0–360°, 1° steps) and optionally mirrored to minimize the circular error against the *real* OP map restricted to oriented channels. This gives one derived OP map per PC pair, and an error (L1: mean absolute circular error; L2: RMSE circular error) between the derived and real map.

Control and significance. For every array we generated 2000 control OP maps by drawing i.i.d. Uniform[0°, 180°) orientations for the oriented channels (unoriented channels fixed at −1), using a deterministic per-array seed. Because the analysis is free to pick whichever of the 21 PC pairs reconstructs the map best, a per-pair significance test is optimistically biased. We therefore used a **min-over-pairs** test throughout: the real statistic is the *minimum* error across all 21 pairs, and the null distribution is built by taking, for *each* control map, its own minimum error across all 21 pairs. The reported percentile is $100 \times \text{mean}(\text{null_min} \leq \text{real_min})$; a percentile below 5% means the real reconstruction beats the control after the multiple-comparison correction. This is substantially stricter than a naive per-pair test and is the only test used for significance claims in this paper.

Code and data availability. All analysis code (`functions_maps.py`, `compute_binsweep.py`, `generate_controls.py`, `repack_binsweep.py`) and parameters (`params_binsweep.yml`) are in the project repository. Figures in this manuscript were generated by the scripts in `paper/` from the saved result tables in `dataframes/`.

3 Results

3.1 Spontaneous activity estimates the OP map in a subset of arrays

Figure 1 shows a representative example (monkey L, array 16): the measured OP map, the geometric map estimated from the best-performing PC pair, and the null distribution of the min-over-pairs

control statistic. The real reconstruction ($L2 \text{ error} = 32.3^\circ$) falls far below the control distribution (percentile = 0.35%). Panels C and E make the underlying geometry explicit: the same PC5-vs-PC6 projection of the spontaneous-activity scores, colored by the measured OP (C) and by the OP the geometric readout assigns (E) — panel E is clean by construction (it *is* the rotation/flip fit), and the question the whole method answers is how much of that same angular structure panel C already carries.

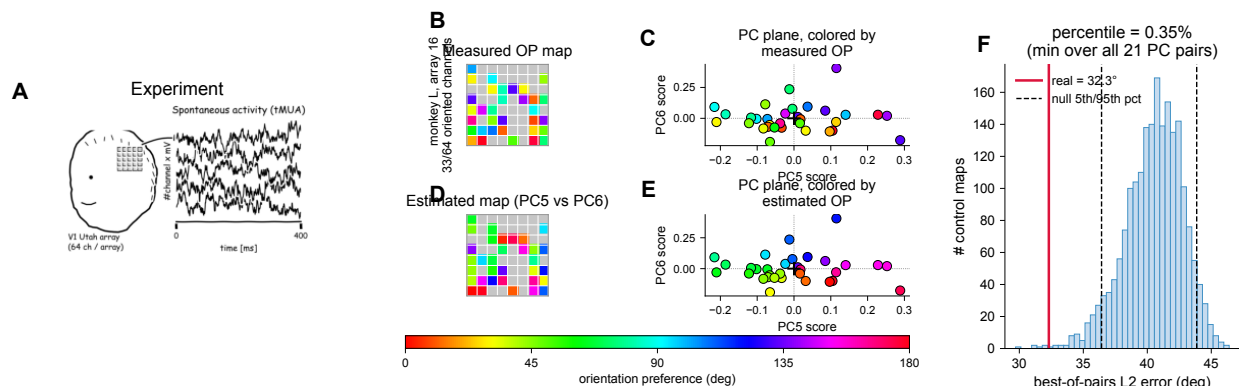


Figure 1: Worked example of OP-map estimation, with the underlying PC-plane geometry made explicit. (A) Schematic of the experiment: tMUA is recorded from a 64-channel V1 Utah array during spontaneous (eyes-closed / eyes-open) activity; a few representative channel traces are sketched for illustration. (B) Measured OP map for monkey L, array 16 (33/64 channels pass the orientation-selectivity criterion; grey = excluded). (C) The same oriented channels plotted in the PC5-vs-PC6 plane (the best-performing pair for this cell; “+” = centroid), colored by their *measured* OP. (D) Geometric map estimated from spontaneous activity using that PC pair. (E) The same PC-plane coordinates, now colored by the *estimated* OP — the clean radial gradient follows directly from the polar-angle rotation/flip fit to panel C. (F) Null distribution of the min-over-pairs control statistic (2000 uniform-random control maps, each reduced to its own best-of-21-pairs L2 error); the real statistic (red) falls at the 0.35th percentile.

This is not a global effect. Figure 2 shows the min-over-pairs percentile for every array, averaged across recording days, separately for the pooled (“all”), eyes-closed, and eyes-open conditions. A subset of arrays is consistently below the 5% threshold across all three conditions — most clearly L13, L14, and L16, and N1 and N6 — while others (e.g. L4, L6, L15, N2, and every array in monkey F) sit well within the null range in every condition.

Turning this into a single per-array **estimability score** — the fraction of (day $\times$ condition) cells with percentile $< 5\%$ — makes the array-specific pattern explicit (Fig. 3A): 6 of 29 arrays are estimated in a majority of their cells (L13, L14, L16, N1, N3, N6), and 16 of 29 arrays (55%) show at least one significant cell. No array in monkey F reached the majority-of-cells criterion, although 3 of the 8 F arrays showed at least one significant cell.

3.2 Estimation requires enough oriented channels

Figure 3B relates the per-cell percentile to the number of oriented channels in the array. Significant cells were essentially absent below roughly 15 oriented channels (10th percentile of oriented-channel count among significant cells = 15), and the fraction of significant cells rises with channel count across the range sampled here. Channel count is not sufficient on its own — several arrays with more oriented channels than the typical estimable array never reach significance in any cell (e.g. F4, 36 oriented channels; L12, 28 oriented channels; L6, 24 oriented channels) — but it is close

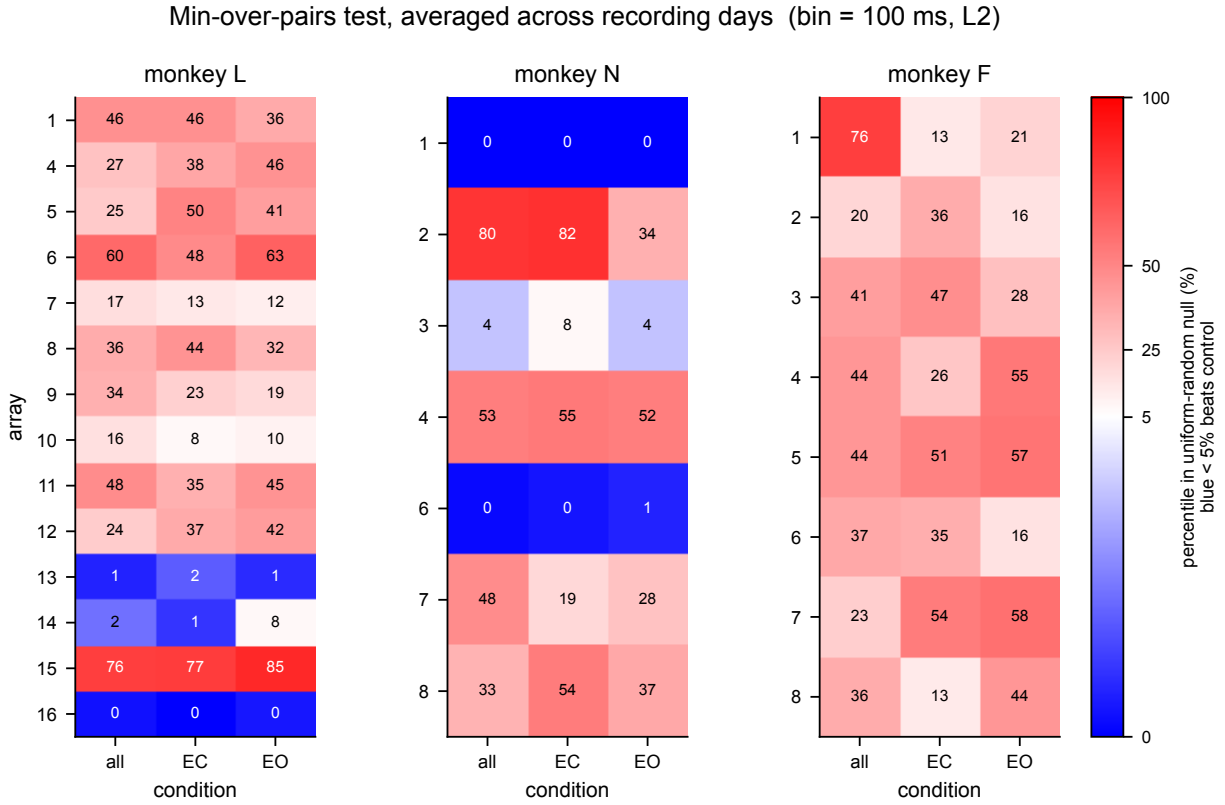


Figure 2: Estimation is array-specific. Min-over-pairs percentile (uniform-random null; blue < 5% beats control), one grid per monkey, rows = array, columns = condition (pooled / eyes-closed / eyes-open), averaged across recording days. Bin = 100 ms, L2 norm.

to necessary: too few oriented channels make the null and the real distributions indistinguishable regardless of whether the array’s map is, in principle, estimable.

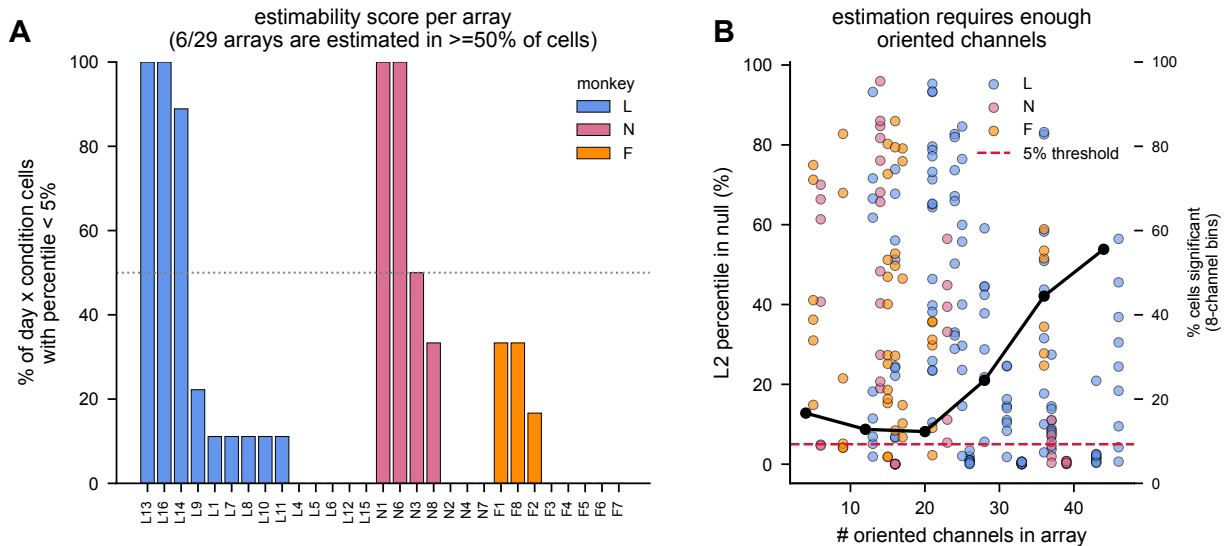


Figure 3: A single estimability score per array, and its dependence on oriented-channel count. (A) Fraction of (day $\times$ condition) cells per array with percentile $< 5\%$, grouped and colored by monkey; 6/29 arrays are estimated in a majority of cells. (B) Per-cell percentile vs. number of oriented channels in the array (colored by monkey); overlaid black line: fraction of cells significant within 8-channel bins of oriented count.

3.3 Estimation is reproducible across independent recording days

The single strongest piece of evidence that this is a genuine effect rather than a selection artifact is that **the same arrays are estimated well on different recording days, each with an independently redrawn control** (Fig. 2 shows this averaged over days; the per-day grids in paper/tables/ show it is not driven by a single day). Chance estimation would not be expected to repeat in the same arrays across sessions collected weeks apart.

Among arrays with enough oriented channels to evaluate reliably (≥ 25 oriented channels, condition = all; 12 arrays across the 3 monkeys), Figure 4 quantifies how similar the *profile* of which PC pairs carry information is from day to day: the median L2 distance between per-pair error profiles was 3.5° (L), 3.4° (N), and 2.6° (F, $n = 1$ array), and the median top-3 winning-pair Jaccard overlap was 0.20–0.35. With only 3 days for monkey L and 2 for monkeys N/F, we treat this descriptively rather than claiming a formal significance level, per the small day count.

Panels C–D of Fig. 4 make this concrete for two representative, fully estimable arrays (L13, L16): on every one of the 3 recording days, a cluster of several PC pairs stays below the 5% threshold, but the single best-performing pair (starred) is a different member of that cluster each day. So day-to-day reproducibility holds at the level of “the map is carried by one of a handful of pairs,” not at the level of a single fixed pair — the summary statistics in panels A–B are one way of quantifying exactly this.

Day-to-day reproducibility of the map-carrying PC pairs
(arrays with ≥ 25 oriented channels, condition = all, bin = 100 ms, L2)

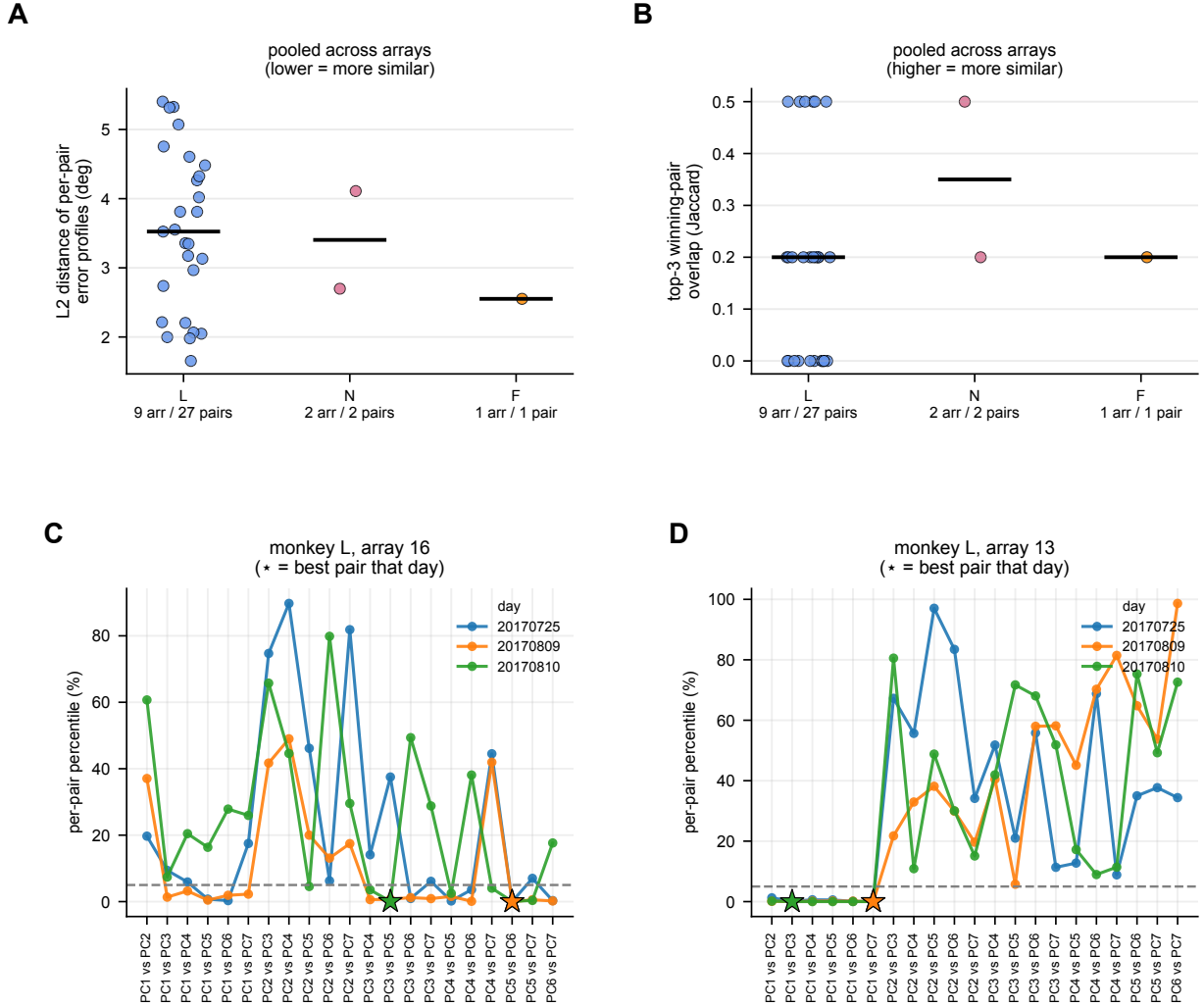


Figure 4: Day-to-day reproducibility of the map-carrying PC pairs, for arrays with ≥ 25 oriented channels evaluated on more than one recording day. (A) L2 distance between the 21-pair error profile on two different days (lower = more similar), pooled across all qualifying arrays. (B) Jaccard overlap of the top-3 best-performing pairs between the same two days (higher = more similar), pooled across arrays. Black bar = median; points = individual day-pairs. (C, D) Concrete per-array illustration for two representative, fully estimable example arrays (monkey L, arrays 16 and 13): the min-over-pairs percentile of each of the 21 PC pairs, plotted separately for each of the array's 3 recording days (colored lines); the star marks the single best-performing pair on that day. A cluster of pairs stays below the 5% threshold (dashed line) on every day, but which specific pair within that cluster wins shifts from day to day.

3.4 The map-carrying principal-component pair is not fixed

Across the cells that were significant in at least one of EC or EO ($n = 23$; Fig. 5B), no single PC pair dominates: roughly a third of the 21 pairs are each significant in 8–10 of the 23 cells, with the rest spread more thinly. This means the “readout direction” through which orientation information appears in the correlation structure shifts across arrays, days, and states — consistent with the hypothesis (to be tested with a model, see Discussion) that changing input or brain state rotates which PCs the map loads onto, rather than the map information disappearing.

3.5 Eyes-closed vs eyes-open estimation is correlated but not identical

Figure 5A compares the min-over-pairs percentile in EC vs EO for every array/day with both conditions available ($n = 72$ cells). Percentiles are positively correlated (points cluster around the diagonal), and 14 cells are significant in both conditions. But EC and EO are not interchangeable: 5 cells were significant in EC only and 4 in EO only — a roughly balanced, mixed picture rather than a systematic advantage for either eye state.

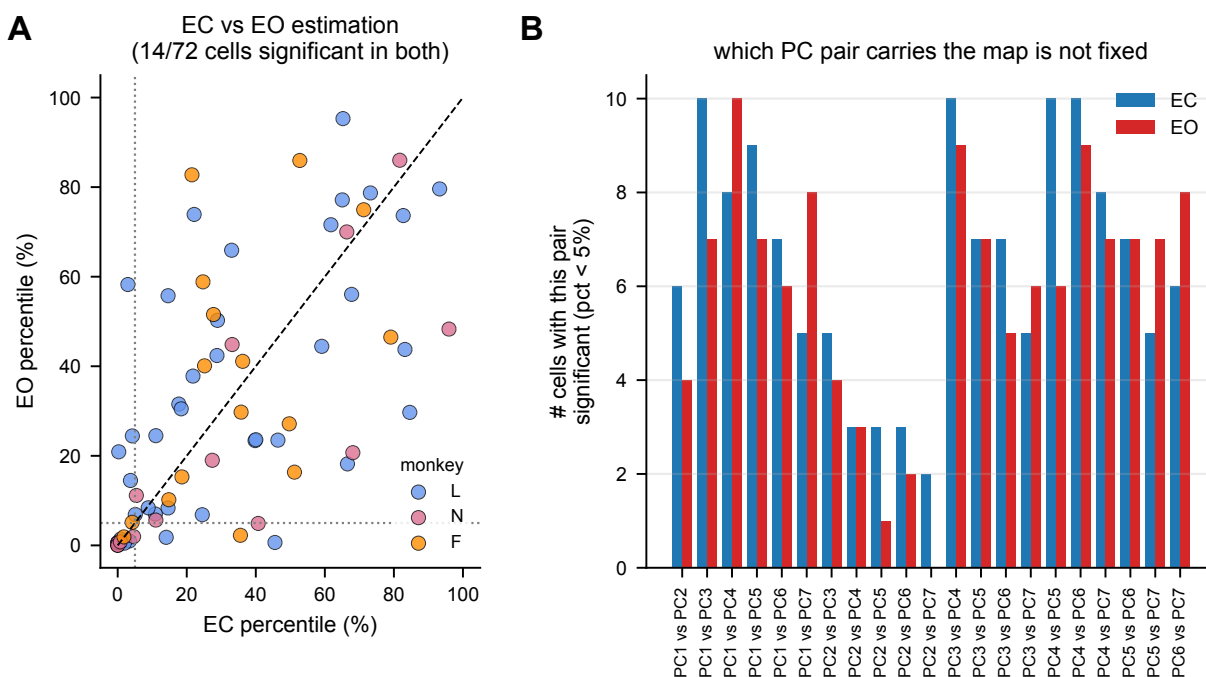


Figure 5: Eyes-closed vs. eyes-open. (A) Min-over-pairs percentile in EC vs. EO for every array/day with both conditions available, colored by monkey; dotted lines mark the 5% threshold. (B) Among cells significant in EC and/or EO, the number of cells for which each of the 21 PC pairs itself reaches the 5% per-pair threshold, split by condition — no single pair dominates.

4 Discussion

(placeholder for full discussion — the following are the load-bearing points from the handover, to be expanded)

- Spontaneous tMUA correlation structure, reduced by PCA and read out with a purely geometric procedure, estimates a previously measured sensory map in roughly a fifth to a half of V1 arrays

(depending on the estimability threshold used), established against a uniform-random control with a selection-corrected significance test.

- Estimation is not uniform across arrays, and the arrays that are estimated well do so reproducibly across independent recording days — the most defensible part of the finding, because chance estimation would not be expected to repeat in the same arrays.
- The instability of *which* PC pair carries the map (Section 3.4) is a natural target for the reviewer objection “an unstable winner is noise.” The defence used throughout this paper is that the null is *also* min-over-pairs (so the correction already accounts for the winner moving around within a single array/day), and that the estimable arrays are still significant after that correction, and reproducibly so across days. A model in which changing input/state rotates which PCs the map loads onto (mentioned in the handover as reproducible by Katarina’s model) would turn this from a weakness into a mechanistic result if included in a revision.
- Two natural next steps, not yet in this draft: (1) regressing the per-array estimability score on a spatial-smoothness measure of the real OP map (Moran’s I) in addition to oriented-channel count, to see whether smoothness of the ground-truth map — not just how many channels are oriented — predicts which arrays are estimated well; (2) the up-state replay observation (spontaneous up-states preferentially reactivating one orientation), which is preliminary and needs a proper up-state detector and a null that accounts for the array’s own OP composition before it can support a claim.

Supplementary figure

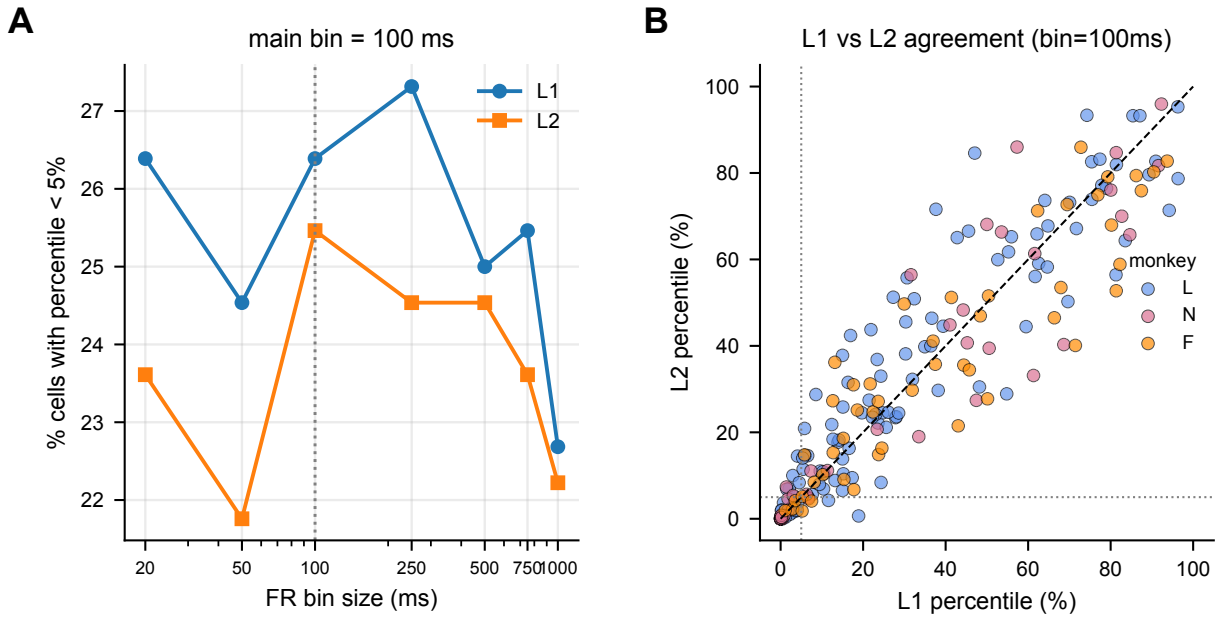


Figure S1: Bin-size and norm robustness. (A) Fraction of cells significant (percentile < 5%) as a function of the firing-rate bin size used to build the correlation matrix, for both error norms; 100 ms (dotted line) was used throughout the main figures. (B) L1 vs. L2 percentile for every cell at the main bin, colored by monkey — the two norms agree closely.